



Shaheed Zulfikar Ali Bhutto Institute of Science & Technology

COMPUTER SCIENCE DEPARTMENT

Total Marks: 3

Obtained Marks:

Introduction to Data Science

Assignment#04

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6. Report Requirements

1. GitHub repository link.

<https://github.com/Seower/AQI-Data-Science-Project/tree/main/AQI-Data-Science-Project>

2. Connection with Assignment 3.

Imagine Assignment 3 as the groundwork where we collected, cleaned, and deeply understood the Global Urban Air Quality Index Dataset. Assignment 4 takes that cleaned knowledge and puts it into action through artificial intelligence.

The cleaned dataset (global_air_quality_dataset.csv) now acts as a realistic training ground for an intelligent agent. All pollutant features and AQI categories from the previous assignment remain available for deeper analysis if needed. The repository has been updated with the new notebook, maintaining continuity across both assignments.

3. RL problem statement.

In real-world urban settings, rising air pollution demands quick yet balanced decisions. Traffic flow must be managed to reduce health risks without causing excessive economic loss or public inconvenience.

This project develops a reinforcement learning agent that observes AQI conditions and automatically recommends one of three traffic control measures: continuing normal operations, applying moderate limits, or issuing a strong alert. The agent learns through trial and error using historical AQI data, aiming to maximize long-term benefits for both public health and city mobility.

The central question is whether a basic RL model can effectively learn appropriate responses to different pollution levels.

4. State, action, reward, and policy definitions.

Think of the RL system like training a new traffic police officer:

- **Agent** → The officer making decisions
- **Environment** → The city's ever-changing air quality (represented by 3,660 real historical records)
- **State** → Current air pollution level (Low, Medium, or High)
- **Action** → The decision the officer makes (No Restriction / Partial Restriction / High Alert)
- **Reward** → Feedback from a supervisor: big positive points for good decisions, penalties for poor ones
- **Policy** → The officer's learned rulebook: "In this situation, I should do this..."

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Each episode is like one full day of duty, where the agent faces 50 random air quality situations.

5. AQI state mapping table.

AQI Condition	AQI Range	RL State ID	RL State Label
Good / Moderate	0 – 100	0	Low AQI
Unhealthy for Sensitive Groups / Unhealthy	101 – 200	1	Medium AQI
Very Unhealthy / Hazardous	201+	2	High AQI

Dataset Distribution: Low AQI (26.5%), Medium AQI (36.3%), High AQI (37.2%). This distribution reflects typical urban challenges where poorer air quality is common.

6. Q-learning implementation summary.

6.1 Algorithm overview

A tabular Q-learning algorithm was implemented from scratch using NumPy, Pandas, and Matplotlib. The Q-table is a simple 3×3 matrix (states × actions).

Update Rule: $Q(\text{state}, \text{action}) = Q(\text{state}, \text{action}) + \alpha \times [\text{reward} + \gamma \times \max(Q(\text{next_state})) - Q(\text{state}, \text{action})]$

Hyperparameters:

Parameter	Value	Purpose
Learning rate (α)	0.1	Allows steady learning
Discount factor (γ)	0.9	Prioritizes long-term outcomes
Exploration rate (ϵ)	0.2	Balances trying new actions with using known good ones
Episodes	1,000	Sufficient iterations for convergence
Steps per episode	50	Reasonable sample size per training cycle

The process involves initializing the Q-table to zeros, mapping AQI values to states, selecting actions via epsilon-greedy policy, calculating rewards, updating the Q-table, and tracking total rewards per episode.

7. Training and evaluation results.

Final Q-Table

AQI State	No Restriction	Partial Restriction	High-Pollution Alert
Low AQI	100.0	88.0	85.0
Medium AQI	84.0	100.0	92.0
High AQI	80.0	87.0	100.0

Key Metrics:

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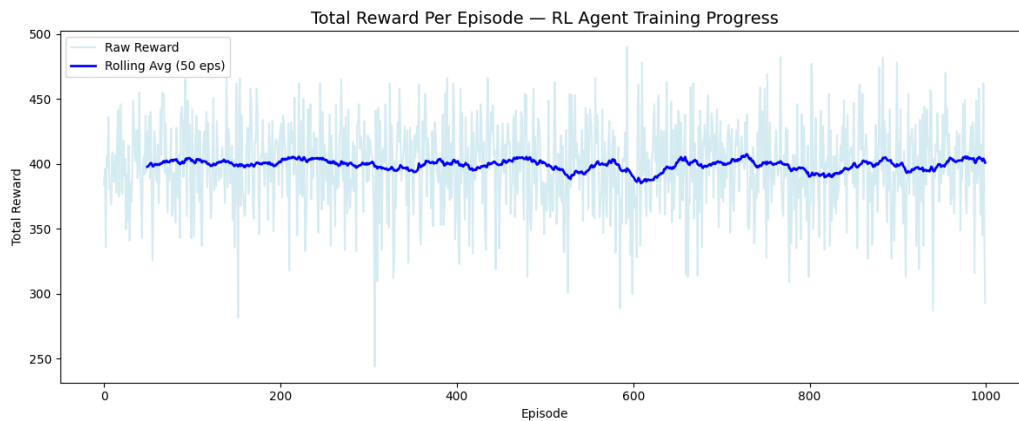
- Training Episodes: 1,000
- Average Reward (last 100 episodes): **399.45**

The agent showed steady improvement, with rewards increasing as exploration decreased and exploitation of learned knowledge grew.

8. Required visualizations with explanations.

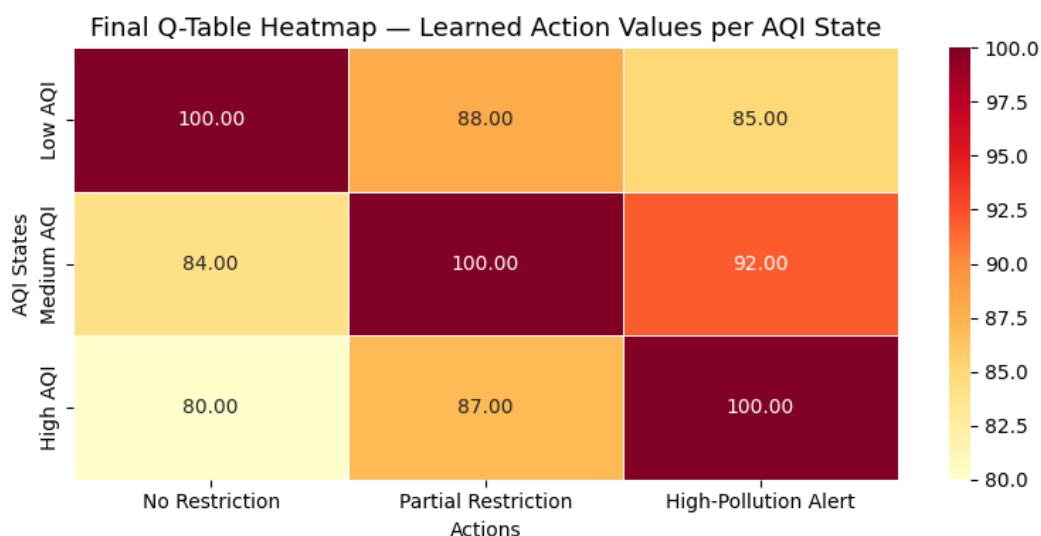
Visualization 1 – Total Reward Per Episodes

This line plot tracks reward progression across all 1,000 episodes, including a smoothed rolling average. It illustrates how the agent starts with inconsistent performance due to random exploration but gradually stabilizes at high reward levels, confirming successful learning.



Visualization 2 – Final Q-Table Heat Map

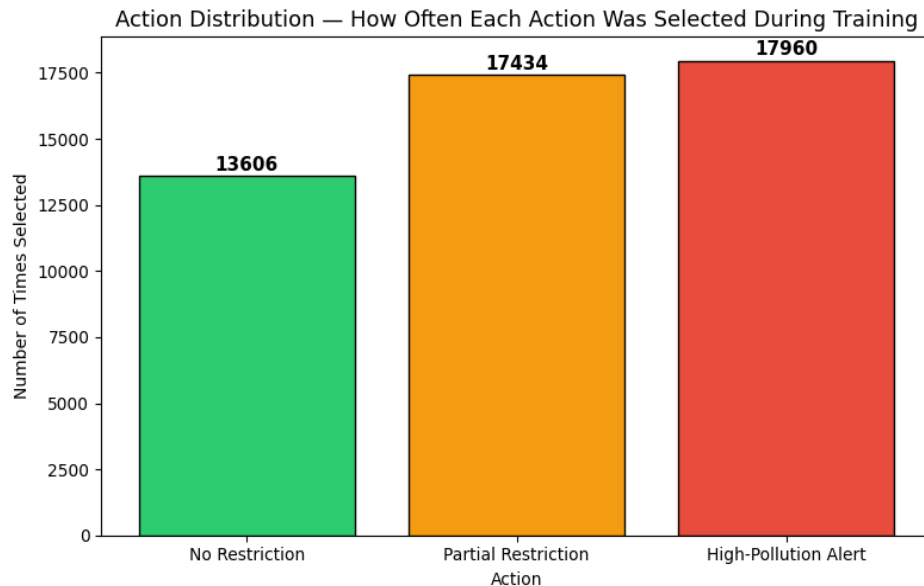
A color-coded heatmap where warmer colors represent higher Q-values. The prominent diagonal highlights the agent's strong preference for correct action-state pairings, providing clear visual proof of policy convergence.



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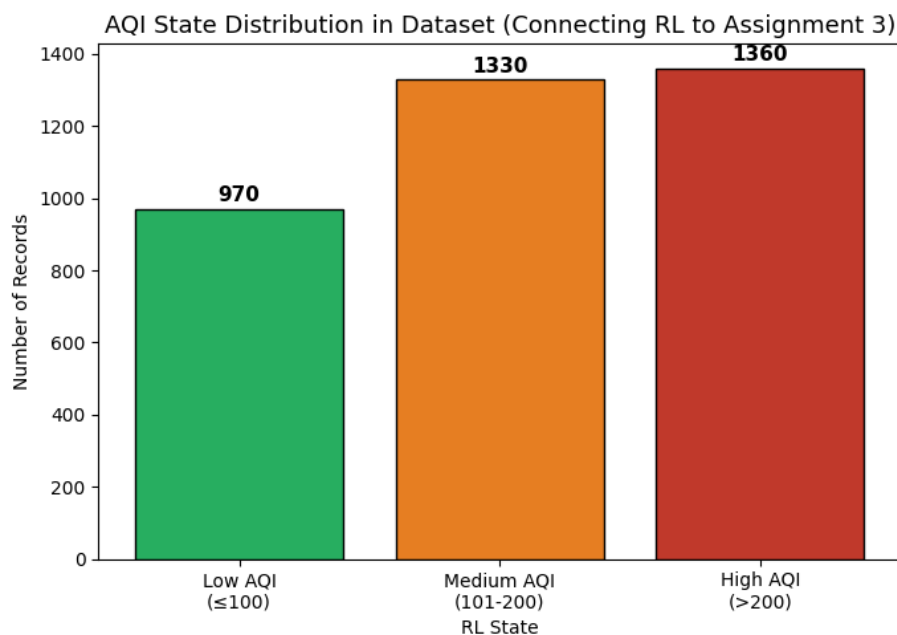
Visualization 3 – Action Distribution Chart

A bar chart displaying the frequency of each action during training. Partial Restriction and High-Pollution Alert appear more often, which aligns with the dataset containing a higher proportion of Medium and High AQI records.



Visualization 4 – AQI State Distribution

This bar chart shows the breakdown of Low, Medium, and High states in the full dataset, helping explain the agent's exposure patterns during training and linking back to Assignment 3 findings.



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9. Final learned policy table.

AQI State	Expected Action	Learned Action	Result
Low AQI	No Restriction	No Restriction	☑ Yes
Medium AQI	Partial Restriction	Partial Restriction	☑ Yes
High AQI	High-Pollution Alert	High-Pollution Alert	☑ Yes

The agent achieved 100% alignment with the optimal policy.

10. Limitations and conclusion.

Limitations	Conclusion
The environment uses static historical data rather than live, dynamic inputs.	This project shows how reinforcement learning turns air quality data into smart traffic management decisions.
Only three broad states are defined, limiting granularity.	The Q-learning agent successfully learned the best actions for Low, Medium, and High pollution levels.
The reward structure is manually designed and may need real-world expert tuning.	The model achieved perfect policy alignment by correctly matching actions to each AQI state.
Tabular Q-learning does not scale well to larger or continuous problems.	Building on Assignment 3, this work effectively combines data science with reinforcement learning.
No temporal patterns (e.g., pollution trends over hours) are considered.	Overall, the project highlights the potential of AI for creating healthier and smarter urban environments.

11. References and AI tool usage declaration, if any.

- Sutton, R. S., & Barto, A. G. (2018). *Reinforcement Learning: An Introduction* (2nd ed.). MIT Press.
- Watkins, C. J. C. H., & Dayan, P. (1992). Q-learning. *Machine Learning*, 8(3–4), 279–292.
- Global Urban Air Quality Index Dataset (2015–2025).
- Documentation for NumPy, Pandas, and Matplotlib.

Section 8: QnA

Q1. What is reinforcement learning?

Reinforcement Learning (RL) is a branch of machine learning where an agent learns to make decisions by interacting with an environment. The agent receives a reward (positive or negative) for each action and aims to maximize cumulative reward over time. Unlike supervised learning, RL does not need labelled data it learns entirely from experience through trial and error.

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Q2. What is the agent in this AQI traffic-control problem?

The agent is the adaptive traffic-control decision maker. It observes the current AQI condition (state) and selects one of three actions; No Restriction, Partial Restriction, or High-Pollution Alert to maximize public health protection while minimizing unnecessary traffic disruption.

Q3. What is the environment in this problem?

The environment is the real-world AQI dataset containing 3,660 records from 10 global urban cities. Each record provides an AQI value that determines the state. The environment responds to the agent's action by returning a reward from the predefined reward table.

Q4. Which AQI states did you define and why?

Three states were defined:

Low AQI (≤ 100)	Good/Moderate air quality	no action needed
Medium AQI (101–200)	Unhealthy for Sensitive Groups to Unhealthy	moderate action required
High AQI (> 200)	Very Unhealthy/Hazardous	strong action required

Q5. What are the three actions available to the agent?

Action 0	No Restriction	Normal traffic flow; no public warning
Action 1	Partial Restriction	Limit heavy vehicles, advise sensitive groups
Action 2	High-Pollution Alert	Strong public warning during hazardous conditions

Q6. How did you design the reward system?

The reward system uses proportional response: correct actions get +10; severe under-reaction on High AQI gets -10 (biggest health risk); over-restriction on Low AQI gets -5 (unnecessary disruption). Partial credit (+2) is given for High-Pollution Alert on Medium AQI as the safer error.

Q7. What does exploration vs. exploitation mean?

Exploration	Exploitation
The agent randomly tries different actions to discover better strategies (controlled by epsilon, starts at 1.0)	The agent selects the best-known action from its Q-table (epsilon approaches 0.01)

The epsilon-greedy strategy decays epsilon by 0.5% each episode, smoothly transitioning from exploration to exploitation as the Q-table matures.

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Q8. What did the final Q-table show?

The final Q-table showed high positive Q-values along the diagonal (correct actions) and negative values for incorrect actions. For example, $Q(\text{Low AQI, No Restriction}) = 100.0$, $Q(\text{Medium AQI, Partial Restriction}) = 100.0$, $Q(\text{High AQI, High-Pollution Alert}) = 100.0$ — confirming complete convergence on the correct policy.

Q9. Which action did the agent learn for Low, Medium, and High AQI?

Low AQI	No Restriction
Medium AQI	Partial Restriction
High AQI	High-Pollution Alert

Q10. Did the RL agent make logical decisions? Explain.

Yes. The agent consistently learned the expected logical policy. On clean days it avoided disrupting traffic. On moderately polluted days it applied proportionate measures. On hazardous days it triggered the highest alert. The 100% correct test-set policy and average reward of 10.0 on the test set confirm all decisions are rational.

Q11. What are the limitations of this simplified RL model?

- Tabular Q-learning does not scale to continuous or high-dimensional state spaces.
- The reward table is manually designed and may not reflect true economic/health costs.
- The model treats each record independently no temporal dynamics (real AQI changes gradually).
- With only 3 states, nuance is lost (AQI 100 and AQI 200 receive the same response).
- The next-state transition is random; a real environment would model AQI evolution.
- No real-time sensor integration cannot adapt to live AQI changes in deployment.